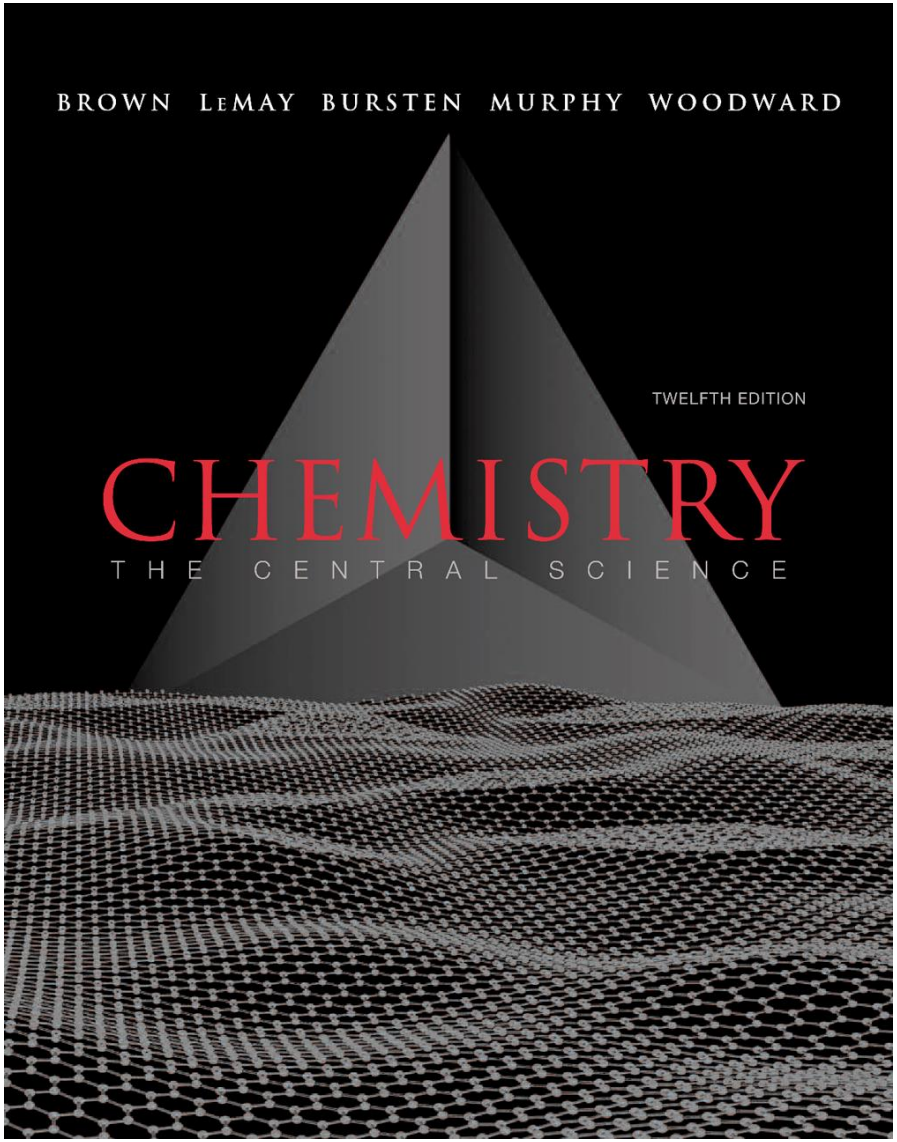


Lecture Presentation

Chapter 16

Acid–Base Equilibria

John D. Bookstaver
St. Charles Community College
Cottleville, MO

© 2012 Pearson Education, Inc.

Some Definitions

- Arrhenius
  - An acid is a substance that, when dissolved in water, increases the concentration of hydrogen ions.
  - A base is a substance that, when dissolved in water, increases the concentration of hydroxide ions.

© 2012 Pearson Education, Inc.

By 1830 it was evident that all acids contain hydrogen but not all hydrogen-containing substances are acids. During the 1880s, the Swedish chemist Svante Arrhenius (1859–1927) defined acids as substances that produce H^+ ions in water and bases as substances that produce OH^- ions in water. Over time the Arrhenius concept of acids and bases came to be stated in the following way:

- An *acid* is a substance that, when dissolved in water, increases the concentration of H^+ ions.
- A *base* is a substance that, when dissolved in water, increases the concentration of OH^- ions.

Hydrogen chloride gas, which is highly soluble in water, is an example of an Arrhenius acid. When it dissolves in water, $HCl(g)$ produces hydrated H^+ and Cl^- ions:

$$HCl(g) \xrightarrow{H_2O} H^+(aq) + Cl^-(aq) \qquad [16.1]$$

The aqueous solution of HCl is known as *hydrochloric acid*. Concentrated hydrochloric acid is about 37% HCl by mass and is 12 *M* in HCl.

Sodium hydroxide is an Arrhenius base. Because NaOH is an ionic compound, it dissociates into Na^+ and OH^- ions when it dissolves in water, thereby increasing the concentration of OH^- ions in the solution.

Some Definitions

- Brønsted–Lowry
  - An acid is a proton donor.
  - A base is a proton acceptor.

© 2012 Pearson Education, Inc.

A Brønsted–Lowry acid…

…must have a removable (acidic) proton.

A Brønsted–Lowry base…

…must have a pair of nonbonding electrons.

© 2012 Pearson Education, Inc.

If it can be either…

…it is **amphiprotic**.

HCO_3^-

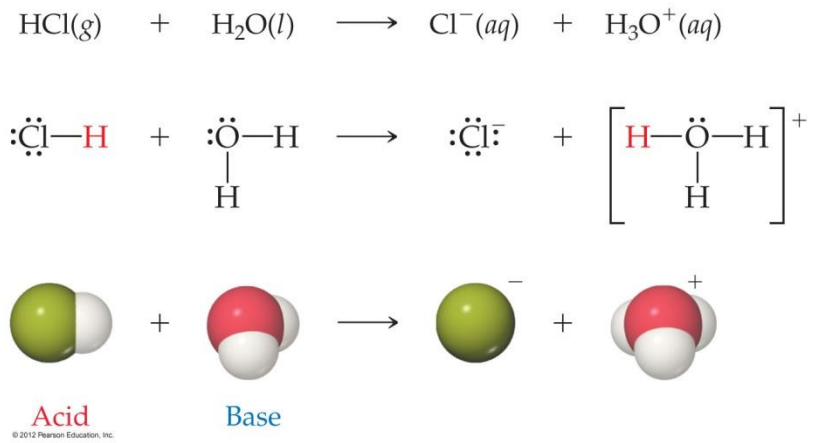
HSO_4^-

H_2O

© 2012 Pearson Education, Inc.

What Happens When an Acid Dissolves in Water?

- Water acts as a Brønsted–Lowry base and abstracts a proton (H^+) from the acid.
- As a result, the **conjugate base** of the acid and a **hydronium ion** are formed.

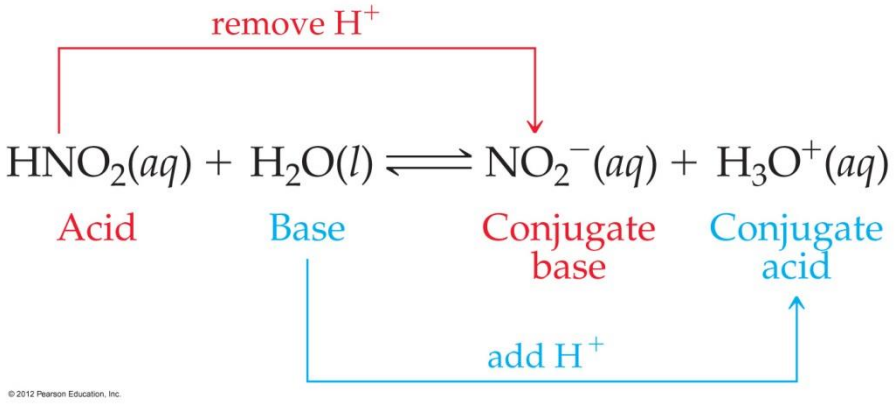


Acids and Bases

© 2012 Pearson Education, Inc.

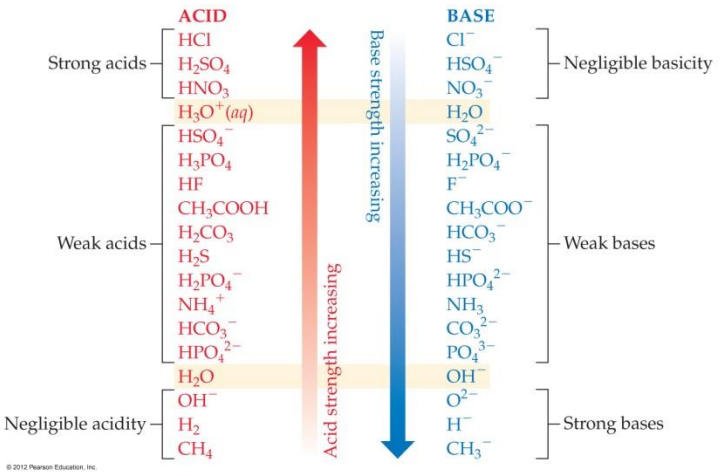
Conjugate Acids and Bases

- The term **conjugate** comes from the Latin word “conjugare,” meaning “to join together.”
- Reactions between acids and bases always yield their conjugate bases and acids.



© 2012 Pearson Education, Inc.

Acid and Base Strength

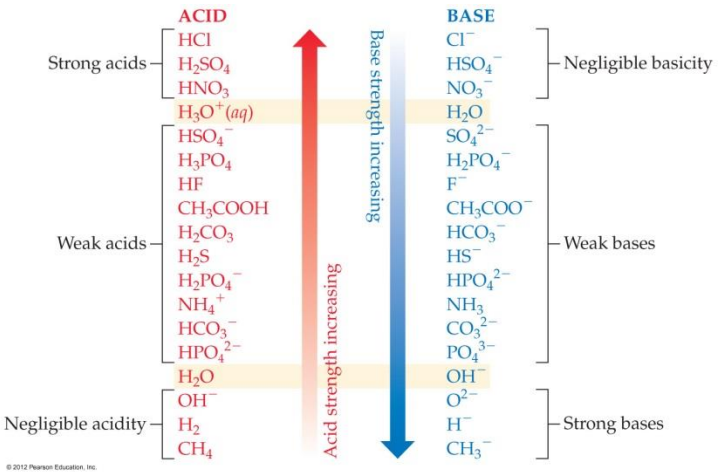


- Strong acids are completely dissociated in water.
  - Their conjugate bases are quite weak.
- Weak acids only dissociate partially in water.
  - Their conjugate bases are weak bases.

Acids and Bases

© 2012 Pearson Education, Inc.

Acid and Base Strength



- Substances with negligible acidity do not dissociate in water.
  - Their conjugate bases are exceedingly strong.

© 2012 Pearson Education, Inc.

Acid and Base Strength

- In any acid–base reaction, the equilibrium will favor the reaction that moves the proton to the stronger base.

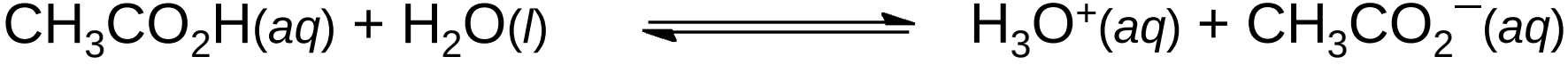
$$HCl(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + Cl^-(aq)$$

- H_2O is a much stronger base than Cl^-, so the equilibrium lies so far to the right that K is not measured ($K >> 1$).

© 2012 Pearson Education, Inc.

Acid and Base Strength

- In any acid–base reaction, the equilibrium will favor the reaction that moves the proton to the stronger base:

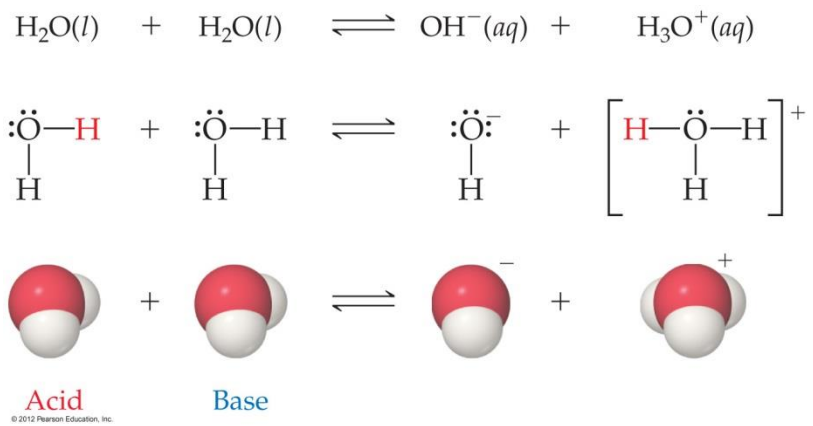


- Acetate is a stronger base than H_2O, so the equilibrium favors the left side ($K < 1$).

© 2012 Pearson Education, Inc.

Autoionization of Water

- As we have seen, water is amphoteric.
- In pure water, a few molecules act as bases and a few act as acids.
- This is referred to as **autoionization**.



© 2012 Pearson Education, Inc.

Ion Product Constant

- The equilibrium expression for this process is

$$K_c = [H_3O^+]\,[OH^-]$$

- This special equilibrium constant is referred to as the **ion product constant** for water, K_w.
- At 25°C, $K_w = 1.0 \times 10^{-14}$

© 2012 Pearson Education, Inc.

pH

pH is defined as the negative base-10 logarithm of the concentration of hydronium ion:

$$pH = -\log [H_3O^+]$$

© 2012 Pearson Education, Inc.

pH

- In pure water,

$$K_w = [H_3O^+]\,[OH^-] = 1.0 \times 10^{-14}$$

- Since in pure water, $[H_3O^+] = [OH^-]$,

$$[H_3O^+] = \sqrt{1.0 \times 10^{-14}} = 1.0 \times 10^{-7}$$

© 2012 Pearson Education, Inc.

pH

- Therefore, in pure water,

$$pH = -\log (1.0 \times 10^{-7}) = 7.00$$

- An acid has a higher $[H_3O^+]$ than pure water, so its pH is <7.
- A base has a lower $[H_3O^+]$ than pure water, so its pH is >7.

TABLE 16.1 • Relationships among (H^+), (OH^-), and pH at 25 °C

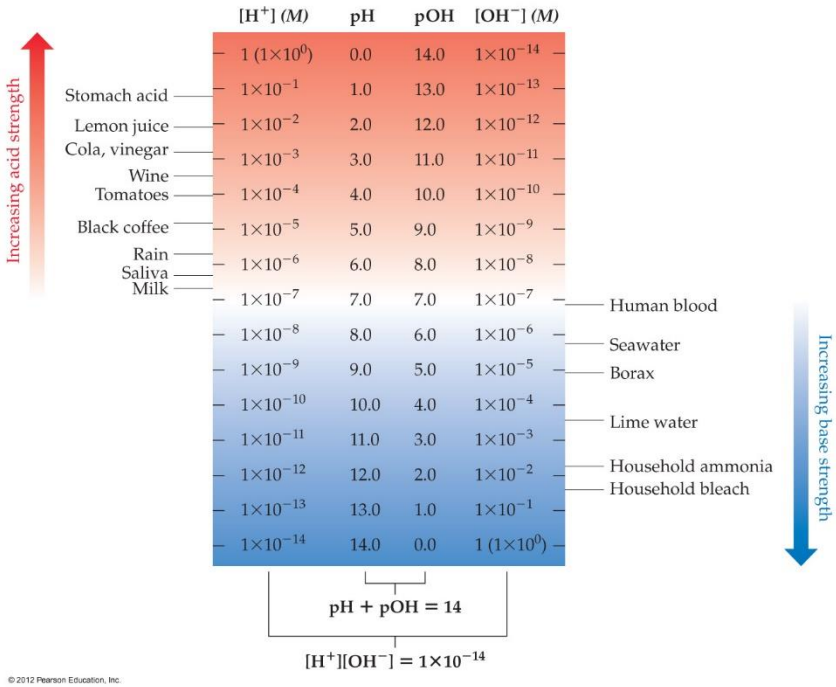
Solution Type	$[H^+]$ (M)	$[OH^-]$ (M)	pH
Acidic	$>1.0 \times 10^{-7}$	$<1.0 \times 10^{-7}$	<7.00
Neutral	1.0×10^{-7}	1.0×10^{-7}	7.00
Basic	$<1.0 \times 10^{-7}$	$>1.0 \times 10^{-7}$	>7.00

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

pH

These are the pH values for several common substances Fig. 16.5.



Acids and Bases

© 2012 Pearson Education, Inc.

Other “p” Scales

- The “p” in pH tells us to take the negative base-10 logarithm of the quantity (in this case, hydronium ions).
- Some similar examples are
  - pOH: $-\log [OH^-]$
  - pK_w: $-\log K_w$

© 2012 Pearson Education, Inc.

Watch This!

Because

$$[H_3O^+]\,[OH^-] = K_w = 1.0 \times 10^{-14}$$

we know that

$$-\log [H_3O^+] + -\log [OH^-] = -\log K_w = 14.00$$

or, in other words,

$$pH + pOH = pK_w = 14.00$$

© 2012 Pearson Education, Inc.

How Do We Measure pH?



Methyl red



Bromthymol blue



Phenolphthalein

© 2012 Pearson Education, Inc.

- For less accurate measurements, one can use
  - Litmus paper
    - "Red" paper turns blue above ~pH = 8
    - "Blue" paper turns red below ~pH = 5
  - Or an indicator.

© 2012 Pearson Education, Inc.

How Do We Measure pH?

For more accurate measurements, one uses a pH meter, which measures the voltage in the solution.



© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Strong Acids

- You will recall that the seven strong acids are HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_3$, and $HClO_4$.
- These are, by definition, strong electrolytes and exist totally as ions in aqueous solution.
- For the monoprotic strong acids,

$$[H_3O^+] = [\text{acid}].$$

© 2012 Pearson Education, Inc.

Strong Bases

- Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca^{2+}, Sr^{2+}, and Ba^{2+}).

- Again, these substances dissociate completely in aqueous solution.

© 2012 Pearson Education, Inc.

Dissociation Constants

- For a generalized acid dissociation,

$$HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

the equilibrium expression would be

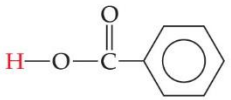
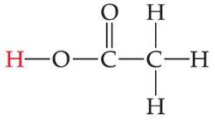
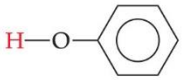
$$K_c = \frac{[H_3O^+]\,[A^-]}{[HA]}$$

- This equilibrium constant is called the **acid-dissociation constant**, K_a.

© 2012 Pearson Education, Inc.

Dissociation Constants

TABLE 16.2 • Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous ($HClO_2$)	H—O—Cl—O	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	H—F	F^-	6.8×10^{-4}
Nitrous (HNO_2)	H—O—N=O	NO_2^-	4.5×10^{-4}
Benzoic (C_6H_5COOH)		$C_6H_5COO^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	H—O—Cl	OCl^-	3.0×10^{-5}
Hydrocyanic (HCN)	H—C≡N	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$C_6H_5O^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

© 2012 Pearson Education, Inc.

The greater the value of K_a, the stronger is the acid.

© 2012 Pearson Education, Inc.

Calculating K_a from the pH

The pH of a 0.10 *M* solution of formic acid, HCOOH, at 25°C is 2.38. Calculate K_a for formic acid at this temperature.

We know that

$$K_a = \frac{[H_3O^+]\,[HCOO^-]}{[HCOOH]}$$

© 2012 Pearson Education, Inc.

Calculating K_a from the pH

The pH of a 0.10 *M* solution of formic acid, HCOOH, at 25°C is 2.38. Calculate K_a for formic acid at this temperature.

To calculate K_a, we need the equilibrium concentrations of all three things.

We can find $[H_3O^+]$, which is the same as $[HCOO^-]$, from the pH.

© 2012 Pearson Education, Inc.

Calculating K_a from the pH

$$\text{pH} = -\log [H_3O^+]$$

$$2.38 = -\log [H_3O^+]$$

$$-2.38 = \log [H_3O^+]$$

$$10^{-2.38} = 10^{\log [H_3O^+]} = [H_3O^+]$$

$$4.2 \times 10^{-3} = [H_3O^+] = [HCOO^-]$$

© 2012 Pearson Education, Inc.

Calculating K_a from pH

Now we can set up a table…

	[HCOOH], *M*	[H_3O^+], *M*	[$HCOO^-$], *M*
Initially	0.10	0	0
Change			
At equilibrium			

© 2012 Pearson Education, Inc.

Calculating K_a from pH

Now we can set up a table…

	[HCOOH], *M*	[H_3O^+], *M*	[$HCOO^-$], *M*
Initially	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
At equilibrium			

© 2012 Pearson Education, Inc.

Calculating K_a from pH

Now we can set up a table…

	[HCOOH], *M*	[H_3O^+], *M*	[$HCOO^-$], *M*
Initially	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
At equilibrium	$0.10 - 4.2 \times 10^{-3}$ $= 0.0958 = 0.10$	4.2×10^{-3}	4.2×10^{-3}

© 2012 Pearson Education, Inc.

Calculating K_a from pH

$$K_a = \frac{[4.2 \times 10^{-3}]\,[4.2 \times 10^{-3}]}{[0.10]}$$

$$= 1.8 \times 10^{-4}$$

© 2012 Pearson Education, Inc.

Calculating Percent Ionization

- Percent ionization = $\frac{[H_3O^+]_{eq}}{[HA]_{initial}} \times 100$
- In this example,

$$[H_3O^+]_{eq} = 4.2 \times 10^{-3}\ M$$

$$[HCOOH]_{initial} = 0.10\ M$$

$$\text{Percent ionization} = \frac{4.2 \times 10^{-3}}{0.10} \times 100$$

$$= 4.2\%$$

© 2012 Pearson Education, Inc.

Calculating pH from K_a

Calculate the pH of a 0.30 M solution of acetic acid, $HC_2H_3O_2$, at 25°C.

$$HC_2H_3O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_2H_3O_2^-(aq)$$

K_a for acetic acid at 25°C is 1.8×10^{-5}.

© 2012 Pearson Education, Inc.

Calculating pH from K_a

The equilibrium constant expression is

$$K_a = \frac{[H_3O^+]\,[C_2H_3O_2^-]}{[HC_2H_3O_2]}$$

© 2012 Pearson Education, Inc.

Calculating pH from K_a

We next set up a table…

	$[C_2H_3O_2]$, *M*	$[H_3O^+]$, *M*	$[C_2H_3O_2^-]$, *M*
Initially	0.30	0	0
Change			
At equilibrium			

We are assuming that *x* will be very small compared to 0.30 and can, therefore, be ignored.

© 2012 Pearson Education, Inc.

Calculating pH from K_a

We next set up a table…

	$[C_2H_3O_2]$, *M*	$[H_3O^+]$, *M*	$[C_2H_3O_2^-]$, *M*
Initially	0.30	0	0
Change	$-x$	$+x$	$+x$
At equilibrium			

We are assuming that *x* will be very small compared to 0.30 and can, therefore, be ignored.

© 2012 Pearson Education, Inc.

Calculating pH from K_a

We next set up a table…

	$[C_2H_3O_2]$, *M*	$[H_3O^+]$, *M*	$[C_2H_3O_2^-]$, *M*
Initially	0.30	0	0
Change	$-x$	$+x$	$+x$
At equilibrium	$0.30 - x \approx 0.30$	x	x

We are assuming that *x* will be very small compared to 0.30 and can, therefore, be ignored.

© 2012 Pearson Education, Inc.

Calculating pH from K_a

Now,

$$1.8 \times 10^{-5} = \frac{(x)^2}{(0.30)}$$

$$(1.8 \times 10^{-5})\ (0.30) = x^2$$

$$5.4 \times 10^{-6} = x^2$$

$$2.3 \times 10^{-3} = x$$

© 2012 Pearson Education, Inc.

Calculating pH from K_a

$$\begin{aligned} \text{pH} &= -\log [H_3O^+] \\ &= -\log (2.3 \times 10^{-3}) \\ &= 2.64 \end{aligned}$$

© 2012 Pearson Education, Inc.

Polyprotic Acids

Polyprotic acids have more than one acidic proton.

If the difference between the K_a for the first dissociation and subsequent K_a values is 10^3 or more, the pH generally depends *only* on the first dissociation.

TABLE 16.3 • Acid-Dissociation Constants of Some Common Polyprotic Acids

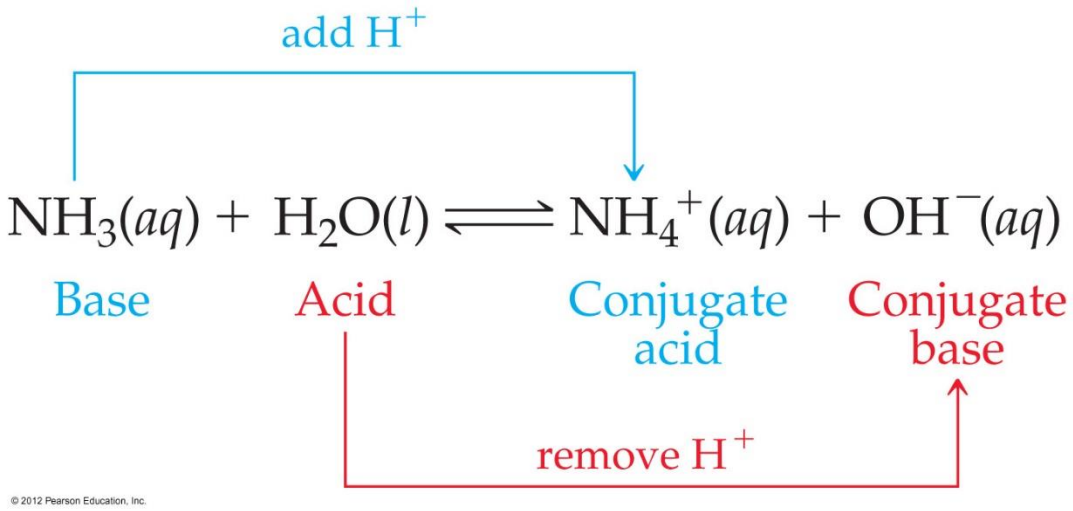
Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$H_2C_6H_6O_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$H_3C_6H_5O_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	$H_2C_2O_4$	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$H_2C_4H_4O_6$	1.0×10^{-3}	4.6×10^{-5}	

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Weak Bases

Bases react with water to produce hydroxide ion.



© 2012 Pearson Education, Inc.

Weak Bases

The equilibrium constant expression for this reaction is

$$K_b = \frac{[HB]\,[OH^-]}{[B^-]}$$

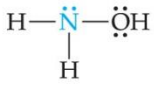
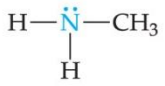
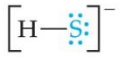
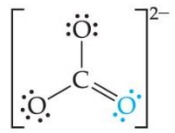
where K_b is the base-dissociation constant.

© 2012 Pearson Education, Inc.

Weak Bases

K_b can be used to find [OH^-] and, through it, pH.

TABLE 16.4 • Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)		NH_4^+	1.8×10^{-5}
Pyridine (C_5H_5N)		$C_5H_5NH^+$	1.7×10^{-9}
Hydroxylamine ($HONH_2$)		$HONH_3^+$	1.1×10^{-8}
Methylamine (CH_3NH_2)		$CH_3NH_3^+$	4.4×10^{-4}
Hydrosulfide ion (HS^-)		H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})		HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)		HClO	3.3×10^{-7}

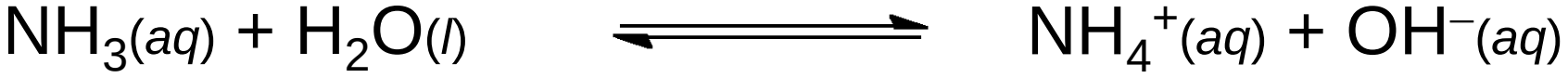
*The atom that accepts the proton is shown in blue.

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

pH of Basic Solutions

What is the pH of a 0.15 *M* solution of NH_3?



$$K_b = \frac{[NH_4^+]\,[OH^-]}{[NH_3]} = 1.8 \times 10^{-5}$$

© 2012 Pearson Education, Inc.

pH of Basic Solutions

Tabulate the data.

	$[NH_3]$, *M*	$[NH_4^+]$, *M*	$[OH^-]$, *M*
Initially	0.15	0	0
At equilibrium			

© 2012 Pearson Education, Inc.

pH of Basic Solutions

Tabulate the data.

	$[NH_3]$, *M*	$[NH_4^+]$, *M*	$[OH^-]$, *M*
Initially	0.15	0	0
At equilibrium	$0.15 - x \approx 0.15$	x	x

© 2012 Pearson Education, Inc.

pH of Basic Solutions

$$1.8 \times 10^{-5} = \frac{(x)^2}{(0.15)}$$

$$(1.8 \times 10^{-5})\,(0.15) = x^2$$

$$2.7 \times 10^{-6} = x^2$$

$$1.6 \times 10^{-3} = x$$

© 2012 Pearson Education, Inc.

pH of Basic Solutions

Therefore,

$$[OH^-] = 1.6 \times 10^{-3}\ M$$

$$pOH = -\log (1.6 \times 10^{-3})$$

$$= 2.80$$

$$pH = 14.00 - 2.80$$

$$= 11.20$$

© 2012 Pearson Education, Inc.

K_a and K_b

TABLE 16.5 • Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO_3	(Strong acid)	NO_3^-	(Negligible basicity)
HF	6.8×10^{-4}	F^-	1.5×10^{-11}
$HC_2H_3O_2$	1.8×10^{-5}	$C_2H_3O_2^-$	5.6×10^{-10}
H_2CO_3	4.3×10^{-7}	HCO_3^-	2.3×10^{-8}
NH_4^+	5.6×10^{-10}	NH_3	1.8×10^{-5}
HCO_3^-	5.6×10^{-11}	CO_3^{2-}	1.8×10^{-4}
OH^-	(Negligible acidity)	O^{2-}	(Strong base)

© 2012 Pearson Education, Inc.

K_a and K_b are related in this way:

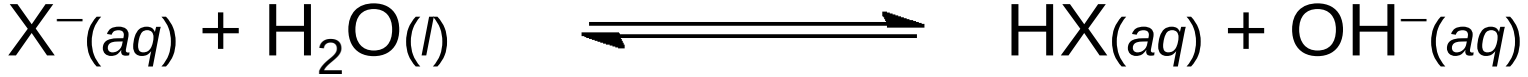
$$K_a \times K_b = K_w$$

Therefore, if you know one of them, you can calculate the other.

© 2012 Pearson Education, Inc.

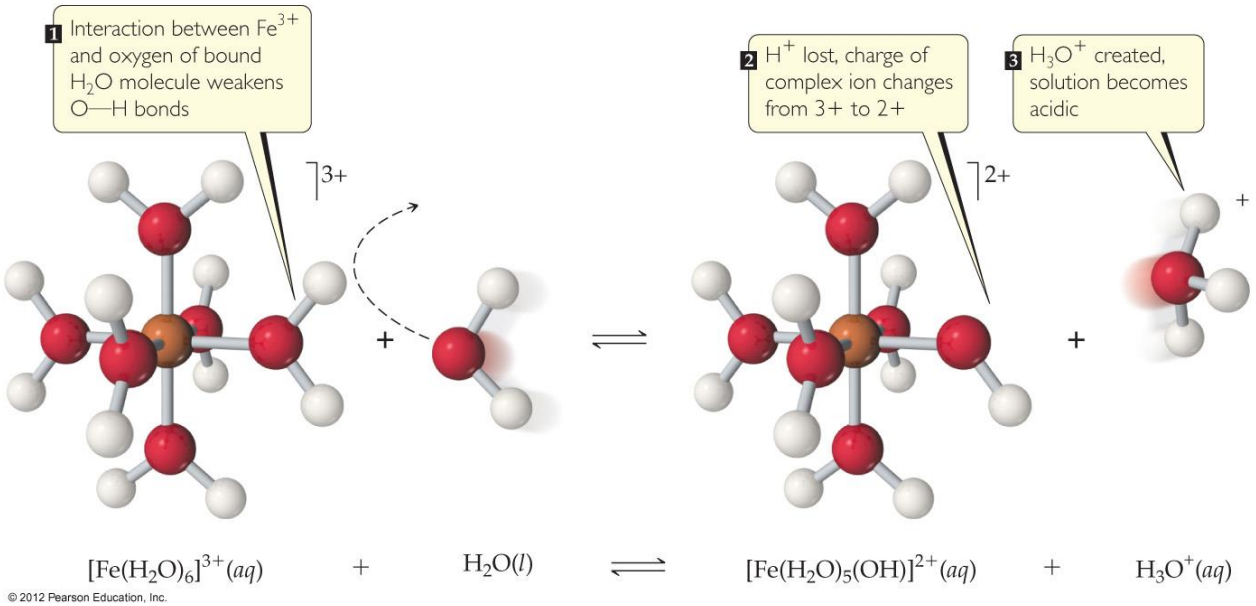
Reactions of Anions with Water

- Anions are bases.
- As such, they can react with water in a **hydrolysis reaction** to form OH^- and the conjugate acid:



© 2012 Pearson Education, Inc.

Reactions of Cations with Water

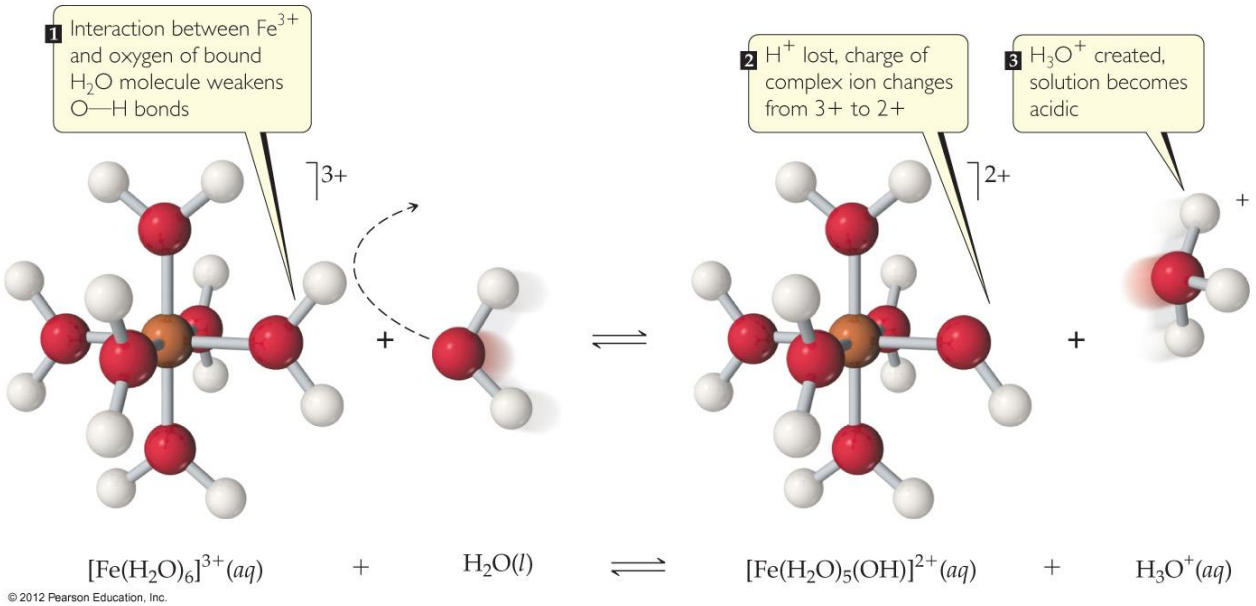


© 2012 Pearson Education, Inc.

- Cations with acidic protons (like NH_4^+) will lower the pH of a solution.
- Most metal cations that are hydrated in solution also lower the pH of the solution.

© 2012 Pearson Education, Inc.

Reactions of Cations with Water



© 2012 Pearson Education, Inc.

- Attraction between nonbonding electrons on oxygen and the metal causes a shift of the electron density in water.
- This makes the O–H bond more polar and the water more acidic.

© 2012 Pearson Education, Inc.

Reactions of Cations with Water

TABLE 16.6 • Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

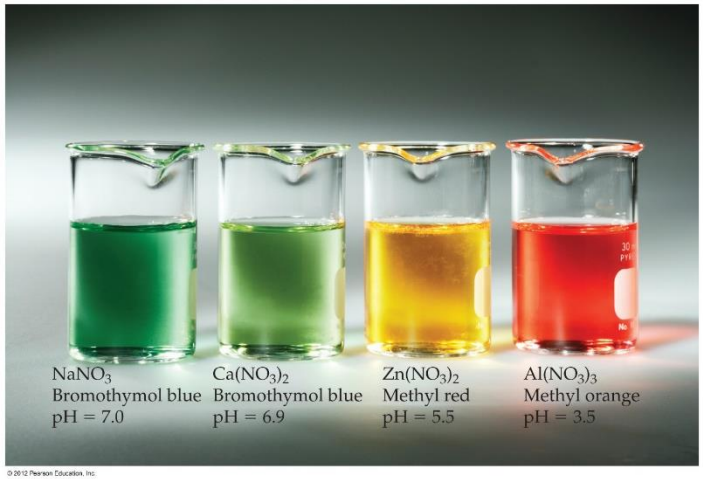
© 2012 Pearson Education, Inc.

- Greater charge and smaller size make a cation more acidic.

© 2012 Pearson Education, Inc.

Effect of Cations and Anions

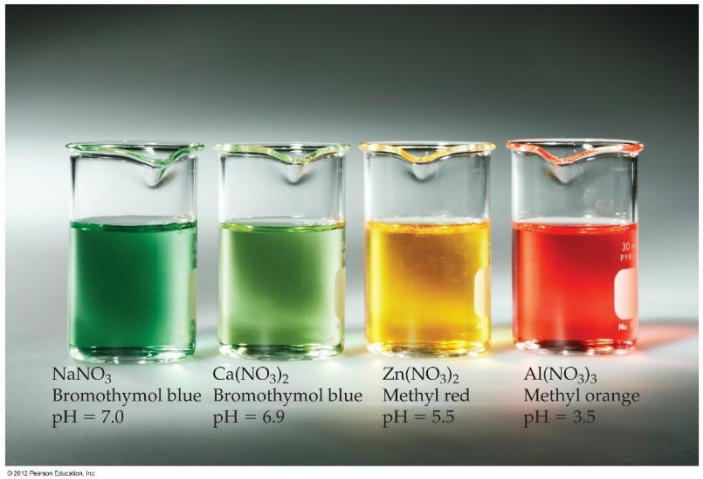
1. An anion that is the conjugate base of a strong acid will not affect the pH.
2. An anion that is the conjugate base of a weak acid will increase the pH.
3. A cation that is the conjugate acid of a weak base will decrease the pH.



© 2012 Pearson Education, Inc.

Effect of Cations and Anions

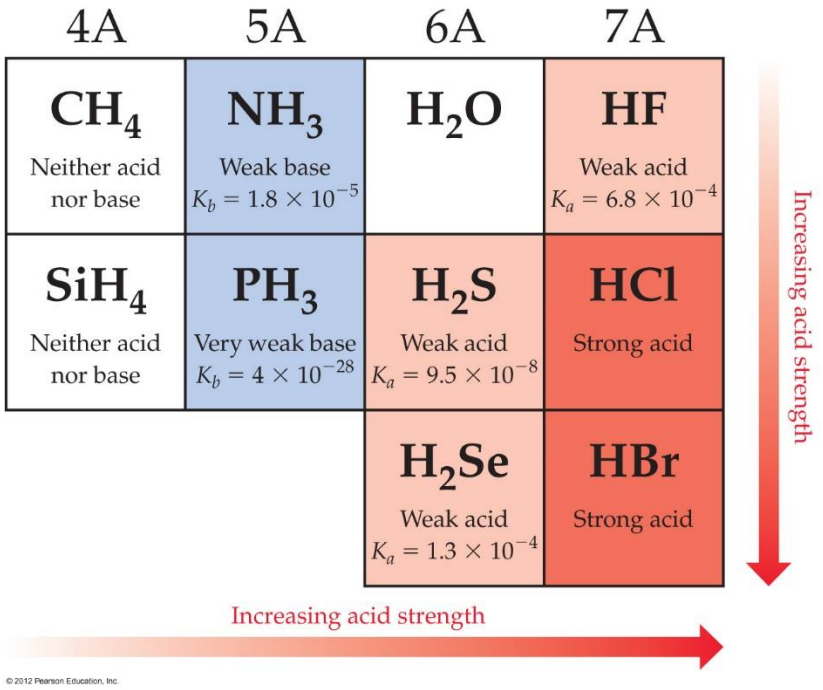
4. Cations of the strong Arrhenius bases will not affect the pH.
5. Other metal ions will cause a decrease in pH.
6. When a solution contains both a weak acid and a weak base, the affect on pH depends on the K_a and K_b values.



© 2012 Pearson Education, Inc.

Factors Affecting Acid Strength

- The more polar the H–X bond and/or the weaker the H–X bond, the more acidic the compound.
- So acidity increases from left to right across a row and from top to bottom down a group.

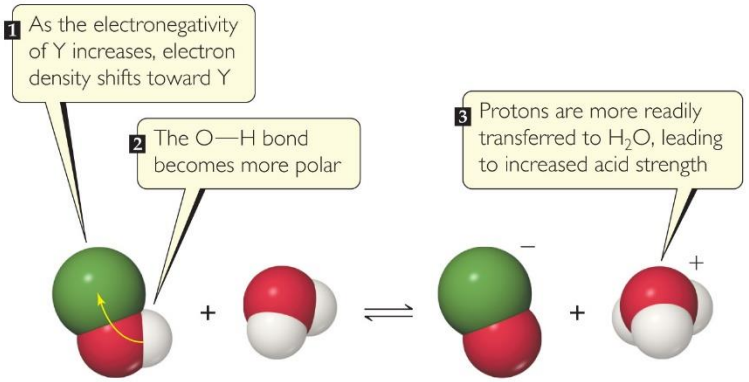


Acids and Bases

© 2012 Pearson Education, Inc.

Factors Affecting Acid Strength

In oxyacids, in which an –OH is bonded to another atom, Y, the more electronegative Y is, the more acidic the acid.

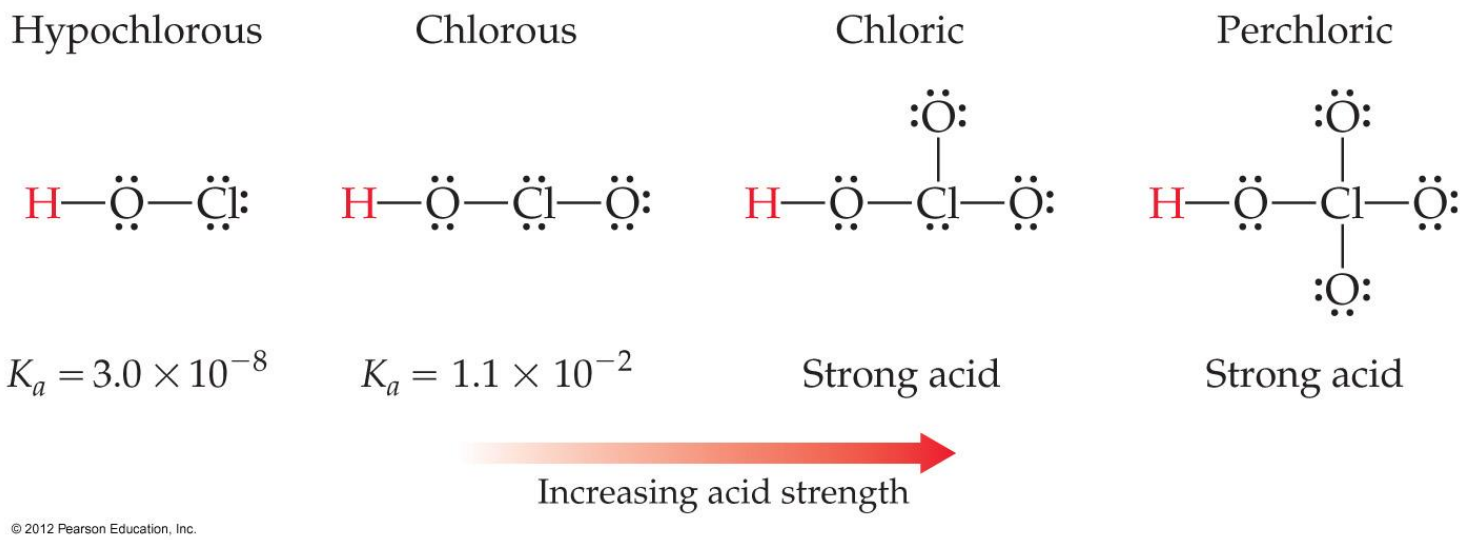


Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

Factors Affecting Acid Strength



© 2012 Pearson Education, Inc.

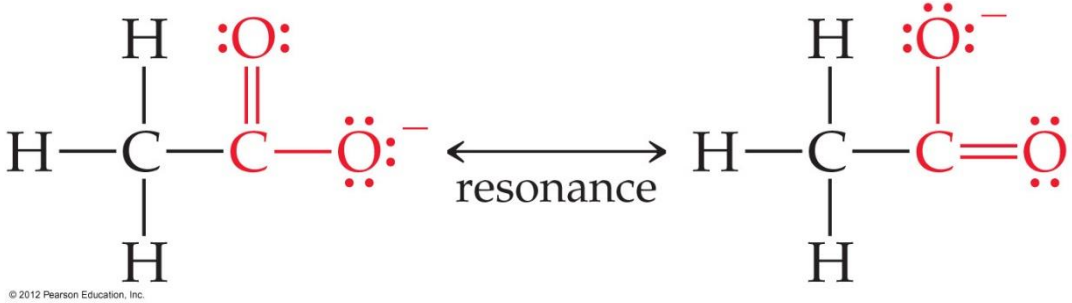
For a series of oxyacids, acidity increases with the number of oxygens.

Acids and Bases

© 2012 Pearson Education, Inc.

Factors Affecting Acid Strength

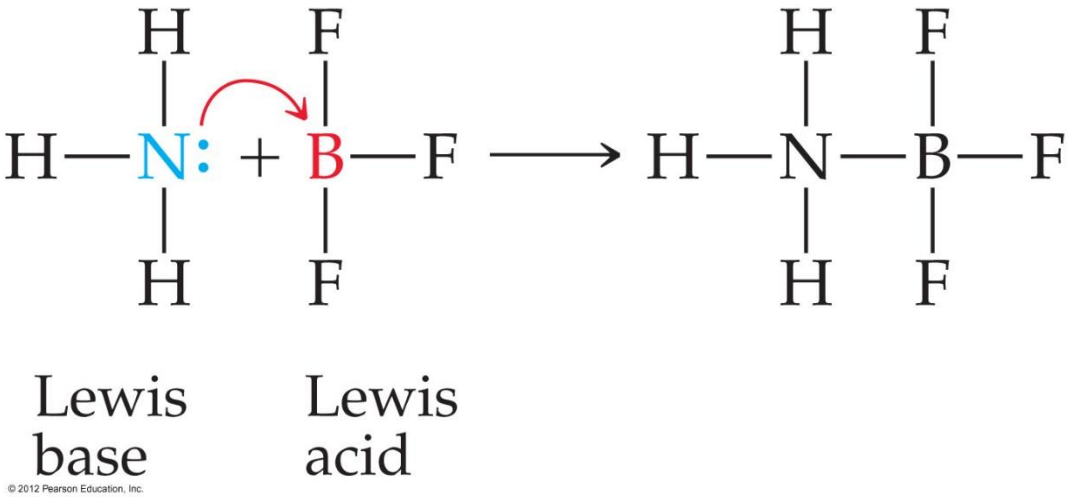
Resonance in the conjugate bases of carboxylic acids stabilizes the base and makes the conjugate acid more acidic.



© 2012 Pearson Education, Inc.

© 2012 Pearson Education, Inc.

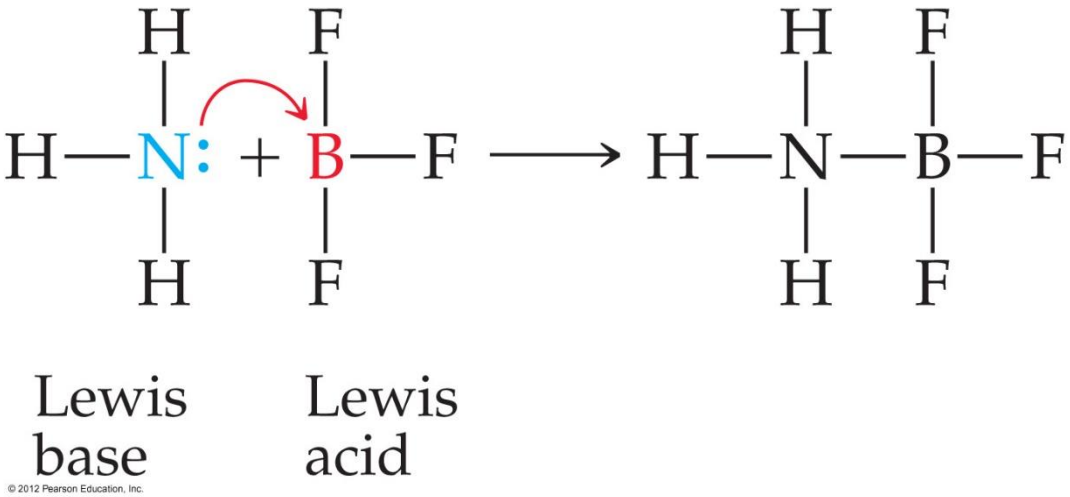
Lewis Acids



- Lewis acids are defined as electron-pair acceptors.
- Atoms with an empty valence orbital can be Lewis acids.

© 2012 Pearson Education, Inc.

Lewis Acids



- Lewis bases are defined as electron-pair donors.
- Anything that could be a Brønsted–Lowry base is a Lewis base.
- Lewis bases can interact with things other than protons, however.

© 2012 Pearson Education, Inc.